

B.Tech Degree V Semester (Supplementary) Examination
June 2011

ME 502 METROLOGY AND MACHINE TOOLS
(2006 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART - A

(Answer ALL questions)

(All questions carry EQUAL marks)

(8 x 5 = 40)

- I. (a) Define the following in the context of Engineering Metrology:
- | | |
|---------------------|-------------------|
| (i) Precision | (ii) Accuracy |
| (iii) Line Standard | (iv) End standard |
| (v) Tolerance | |
- (b) Explain the procedural steps in the manufacture of step gauges.
- (c) With a neat schematic drawing elucidate the application of interferometry in metrology.
- (d) Illustrate the methods of assessment of surface texture.
- (e) Define the following in the context of machine tools:
- | | |
|-----------------|-----------------|
| (i) Reaming | (ii) Trepanning |
| (iii) Broaching | (iv) Turning |
| (v) Shaping | |
- (f) With a neat drawing explain the application of any Quick Return motion mechanism in a shaping machine.
- (g) Explain the following at length with relevant sketches:
- | | |
|----------------------------|----------------------------------|
| (i) upmilling | (ii) downmilling |
| (iii) cylindrical grinding | (iv) continuous path NC machines |
| (v) CMM | |
- (h) Discuss Swiss type automatic machines at length.

PART - B

(4 x 15 = 60)

- II. (a) Differentiate between precision and accuracy. Illustrate what you comprehend by “precise and accurate” and “impressive and inaccurate:.” (5)
- (b) What are the types of fits? Enumerate the various fits with illustrations. (5)
- (c) Give the classification of gauges. (5)
- OR**
- III. (a) Define metrology. Relate accuracy, repeatability and systematic error. What do you comprehend by standardization and interchangeability? (5)
- (b) Illustrate and explain the use of the following:
- | | |
|------------------|------|
| (i) Plug gauges | |
| (ii) Ring gauges | (10) |

(P.T.O.)

- IV. (a) Elucidate the following in the concept of surface finish assessment:
- | | |
|------------------------------------|--------------------|
| (i) Wariness | (ii) lay |
| (iii) CLA | (iv) bearing ratio |
| (v) maximum height of irregularity | |
- (5)
- (b) With a neat schematic drawing explain the principle of operation of the Taylor-Hobson Talysurf. (10)
- OR**
- V. With schematic drawing explain the arrangement of a laser interferometer. (15)
- VI. (a) With neat sketches explain the various work holding devices used in a conventional center lathe. (8)
- (b) Differentiate between Capstan and Turret lathes. (7)
- OR**
- VII. Describe the following machining operations with neat sketches:
- | | |
|-------------------------------|----------------|
| (i) Counter sinking | (ii) Boring |
| (iii) form-tool taper turning | (iv) Annealing |
| (v) T – slot cutting | |
- (15)
- VIII. (a) Explain with sketches gear-holding and indexing in milling machines. } (15)
- (b) Write a short note on numerical control machine tools. }
- OR**
- IX. (a) Distinguish between automatic and semi-automatic machines. } (15)
- (b) Write a short note on NC part programming. }